UNIVERSITY of WASHINGTON

UW Math AI Lab

Spring 2026 · Provable Computation

VERIFIED EXECUTABLE LINEAR ALGEBRA

DenseMatrix makes Lean matrix algorithms run like data.

Row-major data for RREF, LU, and determinant kernels, bridged back to proof-friendly `Matrix` theorems.

POSTER TEAM

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ARTIFACT FILES

Backend Claim

DenseMatrix stores entries in row-major arrays, so matrix code can execute over contiguous data instead of repeatedly re-evaluating pointwise functions.

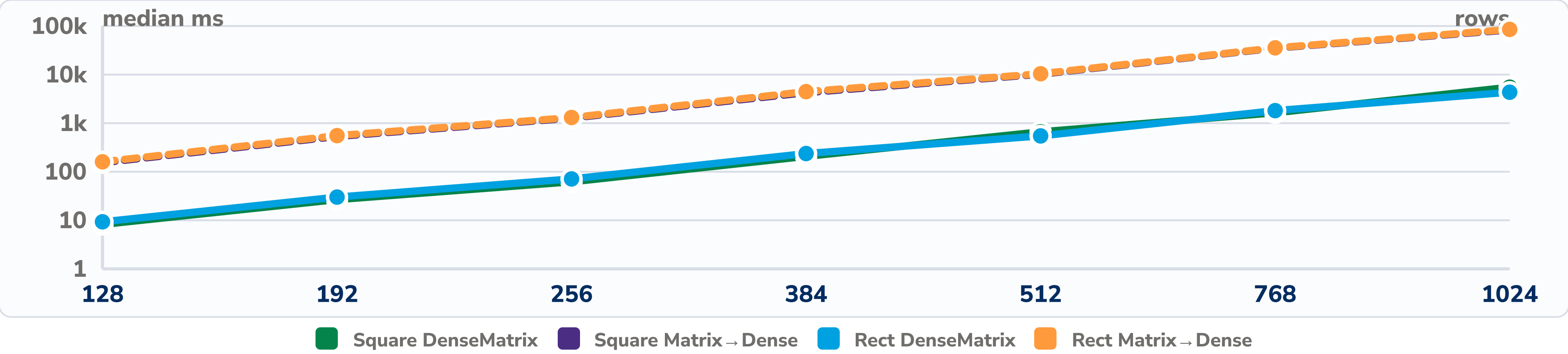
Matrix API → DenseMatrix.ofMatrix → Row-major kernels

Materialized executable workloads only; `Matrix` remains the proof API.

Multiplication Scaling

15.1-21.5x

DenseMatrix vs Matrix→Dense, median ms on a log scale.



Benchmark Scope

2247 RECORDS

328 MATERIALIZED ROWS

2048 MAX ROWS

Protocol	30 repeats · 5 warmups
Sizes	16, 32, 64, 96, 128, 192, 256
Stress	384, 512, 768, 1024
Machine	pinned core · environment archived

Proof Surface

- `DenseMatrix.toMatrix_add`
- `DenseMatrix.toMatrix_mul`
- `rowEchelonForm_toMatrix`
- `luDet_eq_det`
- `gaussDet_eq_det`

Executable backend results stay connected to the mathematical API by theorem anchors.

Representation Tradeoff

Pointwise primitives at the largest complete size.

OPERATION	DENSE	MATRIX	MATRIX→DENSE	WINNER
Add	54.0	45.1	47.5	Matrix
Scalar	53.2	30.3	35.3	Matrix
Transpose	55.3	43.5	46.8	Matrix

```
scripts/build_densematrix_poster_data.mjs --input bench-results/2026sp/<suite>.tar.gz
scripts/export_densematrix_poster_pdf.sh
```

Algorithm Evidence

Dense prebuilt Rat workloads, when available.

REF	n=128	16292 ms
RREF	n=128	32394 ms
LU factor	n=128	17090 ms
Gauss det	n=128	16234 ms
LU det	n=128	16933 ms